Vaned channels and shroud thickness calculations

Index

1. Purpose 3

2. Scope and field of application 3

3. Definitions 3

4. General Procedure - Description 4

5. Description of the methodology 5
   5.1. Dimensioning of the thickness on blades and shroud 5
   5.2. Calculation of impeller and diffuser fatigue stress 7

6. Established Practices 12

7. Description of the module 14

8. Limitations 15

9. References 15

10. Attachments 16
   10.3 Check of the blade thickness AXD 800.600 (project 18C039) 16

1. Purpose

This document establishes the criteria to calculate and to check the thickness of vaned channels and shroud by means of Gulich [1] formulations.

2. Scope and field of application

This procedure is applicable to open and closed impeller and to open and closed diffuser.
This procedure is applicable to the plant of Nova Milanese and plan of La Spezia.

3. Definitions

B2 or **b3**	channel width	mm or in
D2max	maximum external diameter	mm or in
D3 or **Dsw**	diameter of diffuser LE or volute tongue cutwater	mm or in
e	nominal thickness	mm or in
e2	TE thickness	mm or in
eLE	LE thickness	mm or in
em	average thickness	mm or in
eRs	shroud disk thickness	mm or in
Hst.bep	pump head at best efficiency point	m or ft
Hst.max	max head per stage	m or ft
LE	Leading edge of the vane/blade	
n	rotational speed of the pump rotor	rpm
Nq or **Ns**	stage specific speed (n*(Q.bep/suction sides)^0.5/Hst.bep^0.75)	
PS	pressure side of the vane/blade	
Qbep	flowrate at best efficiency point	m^3/h or GPM
rf	fillet radius between disk an vanes/blades	mm or in

Rm	Tensile strength	MPa or lbf/in2
Rp0.2	Yeld strength	MPa or lbf/in2
S.G	fluid specific gravity	
Sbw	Safety factor against fatigue fracture	
SF	global safety factor against blades and shroud rupture	
SS	suction side of the vane/blade	
Sz	Safety factor against blade rupture	
Szz	Safety factor against shroud rupture	
TE	Trailing edge of the vane/blade	
U2	peripheral velocity at external diameter of impeller	m/2 or ft/s
Z	number of vanes or blades	
β2b	impeller exit blade	°
δ	angle between PS and SS of diffuser or volute LE	°
ρματ	material density	kg/m3 or lb/ft3
σbw	Alternate bending strength	MPa or lbf/in2
σm	Mean stress	MPa or lbf/in2
σw	Alternate bending stress	MPa or lbf/in2

4. General Procedure - Description

The procedure described in the document is intended to support the pump designer on the definition of the thickness blades and shrouds to minimize the risk of rupture of pump components, especially impeller or diffuser.

The main parameters that affect the thickness dimensioning are:

- Distance between impeller and diffuser LE or volute cutwaters.
- Casting quality
- Material
- Pumped fluid
- Operative conditions
- Impeller or diffuser dimensions

“Mod.527” spreadsheet is an empirical procedure that should be used in two different ways:

- Definition of minimum recommended thickness of vanes and shroud disks for new design
- Calculation of the safety factor against fatigue rupture of vanes and shroud on existing component

5. Description of the methodology

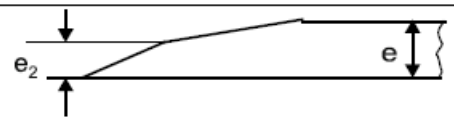
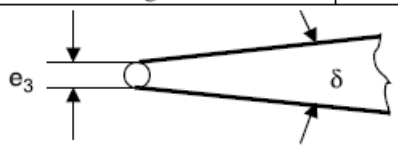
The following sessions describe the methodology used into the “Mod.527”

5.1. Dimensioning of the thickness on blades and shroud

The thickness dimensioning is based on the formulation proposed by [1] on table 14.3. The method evaluates the thickness of blades and shrouds for all centrifugal pumps. On Figure 1 Table 14.3 of [1] summarize the formulations used for the selection of the main vales of thickness, for open or closed impeller or diffuser

In addition to the data on Figure 1 the following assumption must be satisfied:

eLE	min 0.66*e
rf	0.3*e (min 3mm)

Table 14.3 Thickness of blades, vanes and shrouds				Eq.
Closed impellers	Nominal blade thickness e (without profiling)	u_{Ref} = 100 m/s; $n_{q,Ref}$ = 100; z_{Ref} = 7 applicable to z_{La} = 1 to 10		14.3.1
	$e^* \equiv \frac{e}{d_2} = 0.015 \left(\frac{z_{Ref}}{z_{La}}\right)^{0.4} \left(1 + 0.25\left\{\frac{u_2}{u_{Ref}}\right\}^2\right)\left(1 + 0.5\left\{\frac{n_q}{n_{q,Ref}}\right\}^2\right)$			
				
	Trailing edge thickness	$e_2 = e/2$		14.3.2
	Mean blade thickness for calculation according to Table 14.1	$e_m = 0.5(e_2 + e) = 0.75 \times e$		14.3.3
	Shroud thickness	$e_{Rs} = 1.25 \times e$	e from Eq. (14.3.1)	14.3.4
Open impellers	Blade thickness at outer streamline	$e_a = e$		14.3.5
	Blade thickness at inner streamline (rear shroud or hub). Note: make e_i not much larger than e_{Rs} to avoid casting problems.	$e_i \leq 2 \times e_a$		14.3.6
	Mean blade thickness for calculation according to Table 14.1	$e_m = 1.125 \times e$		---
Diffusers	Leading edge thickness	$e_3 = 0.01 \times d_2$		14.3.7
	Mean vane thickness for calculation according to Table 14.1	$e_m = e_3 \times (1 + 5 \tan\delta)$		14.3.8
		δ = angle between suction and pressure side of diffuser vanes: $\delta \geq 5°$		---

- With careful design and manufacturing these recommendations are thought to be on the safe side, but beware of resonances between impeller eigen frequencies and hydraulic or acoustic excitation.
- High-head sea water applications require higher thicknesses.
- The minimum thickness required for the actual casting process must be observed.
- Cast impellers minimum: $e = 0.015 \times d_2$

Figure 1 table of thickness dimensioning as per TABLE 14.3 of [1]

5.2. Calculation of impeller and diffuser fatigue stress

The approach provides a safety margin calculation against fatigue rupture based on formulation proposed by [1], Table 14.1 and 14.2, as shown in Figure 3 and Figure 4.

The following assumptions are applied:

- Blades of closed impeller and diffusers are modelled by beams clamped on the hub and shroud
- The worst operative condition in terms of stresses should be considered during the thickness verification. A typical value is at 0.6Qbep
- Open impeller and diffusers are considered like a beam clamped only on hub.
- The calculation considers the last portion of blades at outlet of the impeller or the first portion of vanes at the inlet of the diffuser as clamped beam with the following dimensions:5*e as width and b2 or b3 as length. If the blade has different thickness on impeller TE or diffuser LE an equivalent value of mean thickness em is used, as shown in Figure 1.
- The blade load on impeller or diffuser at maximum head is used to determine the mean stress. This datum can be calculated from CFD analyses. This term is opposite to centrifugal force so this value is negative when impeller blade thickness is analysed, otherwise it must be considered positive when diffuser blades are investigated. If this term is unknown a rough estimation of the blade load on impeller is based on the following formulation:
  - Dp.bladeload=0.75*Hst(nq/250) for impeller
  - Dp.bladeload=0.1*0.5*ρ*u2^2 for diffuser

Steady forces generate mean stress on the clamped sections

Pressure pulsations generate alternative bending stresses. If this value is unknown a statistical value is proposed, as function of the distance between the impeller outlet and diffuser /volute inlet. This term corresponds to data extracted at 0.6Qbep.

The bending stresses are affected by a notch factor that is function of casting quality and in strictly dependent to the fillet radius between blades and disks.

Notch factor value is:

- 4 or higher for poor casting quality
- 2 or higher for average casting quality

- 1.5 or higher for high quality casting when rf/e>0.5

Higher values of notch factor for each casting class can be achieved by small fillet radii as per formulation in Figure 3.

Safety factor against rupture is the ratio between the Rm and the allowable mean stress. It depends on the material and on topology of the component (blades or shroud). On [1] on Tables 14.11 to 14.16, a list of safety factors for blades Sz and for shrouds Szz is proposed for different materials.

The alternate bending stress σbw is usually unknown. under these circumstances an arbitraty value is calculated using the following correlations:

- σbw = 0.3 Rm for no corrosive fluid (i.e. fresh water or hydrocarbon)
- σbw = 0.12 Rm for corrosive fluid (i.e. sea water)

Safety factor against fatigue fracture Sbw is the ratio between the σbw and the allowable alternate stress. Usual value to consider is Sbw=2

Based on the above assumption a Goodman chart is realized for both blade and shroud, as shown in Figure 2:

- Black line bounds the limit of the material (x=Rm, y= σbw)
- The red line defines the allowable stress region (x=Rm/Sz;y= σbw /Sbw)
- The red point corresponds to the actual stress of the blade (σm; σw)

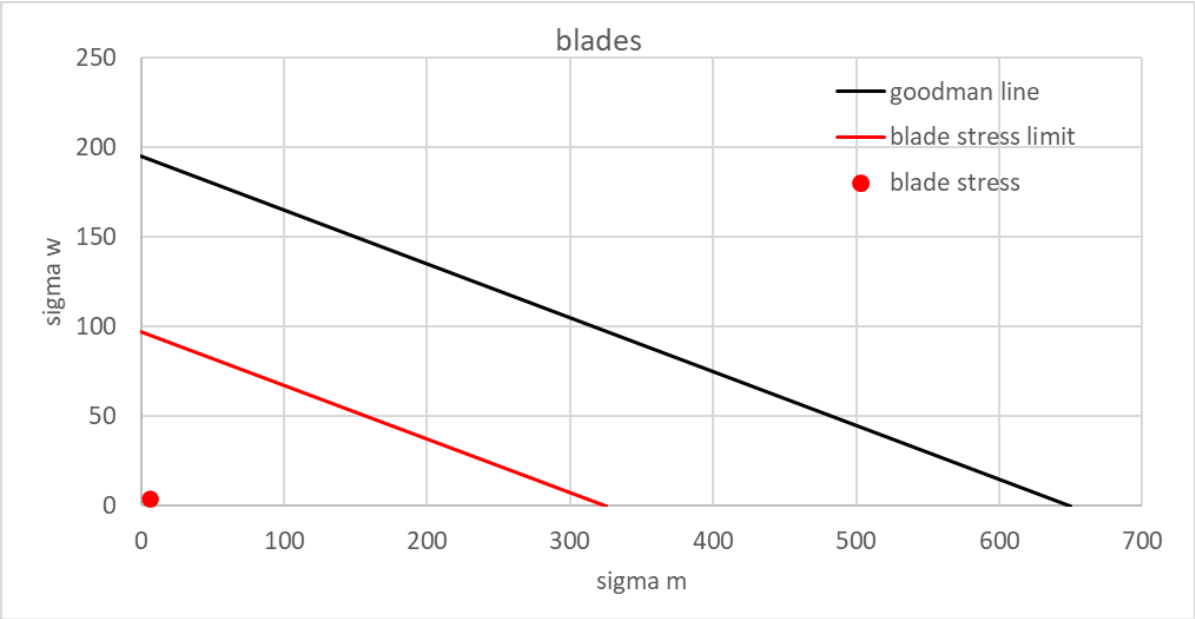
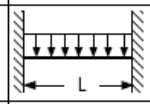
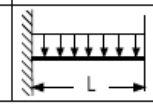
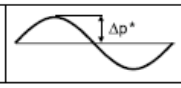
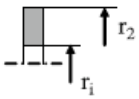


Figure 2. Goodman chart representation.

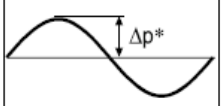
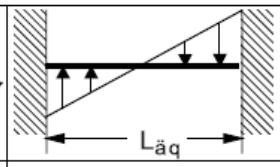
Global safety factor against rupture SF is the ratio between the allowable and the actual stress. The allowable value corresponds to the intersection between the allowable line with the actual stress function ($y=\sigma w/\sigma m*x$)

Table 14.1 Impeller or diffuser vane loading

						Eq.
Bending of a beam by an equivalent uniform loading			diffuser	Open impeller or diffuser		Eq.
	Static or dynamic pressure difference					---
Static vane loading	$\Delta p_{stat} = \frac{\rho}{2} u_2^2 \psi_{load}$		$\psi_{Load} = \frac{2(p_{DS} - p_{SS})}{\rho u_2^2}$			14.1.1
Static load due to centrifugal force	$\Delta p_z = \rho_{Mat} \omega^2 r e \cos\beta_{2B}$					14.1.2
Dynamic vane loading	$\Delta p_{dyn} = \frac{\rho}{2} u_2^2 \Delta p^*$		$\Delta p^* = \frac{0.008}{(d_3^* - 1)^{0.9}}$			14.1.3
Bending moment M, section modulus W, bending stress σ	$M = F_T \frac{x \Delta p L^2}{12}$		$W = \frac{x}{6} e_m^2$	$\sigma = \frac{M}{W} = F_T \frac{\Delta p}{2} \left(\frac{L}{e_m}\right)^2$		14.1.4
	F_T	L	Y		ψ_{Load}	---
Impeller: closed	1	b_2	4	Centrifugal force acting	Set negative	14.1.5
Impeller: open	6					
Diffuser: closed	1	b_3	0	No centrifugal force	Set positive	
Diffuser: open	6					
Impeller blade mean stress: σ_m	$\sigma_m = F_T \frac{\rho}{4} u_2^2 \left(\frac{b_2^*}{e_m^*}\right)^2 \left\{\psi_{Load} + Y e_m^* \frac{\rho_{mat}}{\rho} \cos\beta_{2B}\right\}$					14.1.6
Impeller blade: alternating bending stress σ_w	$\sigma_w = F_T \alpha_k \Delta p^* \frac{\rho}{4} u_2^2 \left(\frac{b_2^*}{e_m^*}\right)^2$		$\alpha_k = 1.2 \left(\frac{e}{r_f}\right)^{0.15}$	r_f = fillet radius between blade and Shroud		14.1.7
Allowable alternating bending stress $\sigma_{bw,al}$	$\sigma_{bw,al} = \frac{\sigma_{bw}}{S_{bw}} \left(1 - \frac{\sigma_m}{R_m} S_z\right)$					14.1.8
Allowable impeller circumferential speed	$u_{2,al} = 2 \frac{e_m^*}{b_2^*} \sqrt{\frac{\sigma_{bw}}{F_T \rho \left\{S_{bw} \alpha_k \Delta p^* + S_z \frac{\sigma_{bw}}{R_m} \left(\psi_{load} + Y e_m^* \frac{\rho_{mat}}{\rho}\right)\right\}}}$					14.1.9
Tangential stress at r_i due to centrifugal force in a disk with a central bore	$\sigma_t = \frac{3+\nu}{4} \rho_{mat} u_2^2 \left\{1 + \frac{1-\nu}{3+\nu} \left(\frac{r_i}{r_2}\right)^2\right\}$					14.1.10

x in Eq (14.1.4) is the width of the "beam" considered; x cancels when calculating the stress. S_{bw} = safety factor against fatigue fracture; S_Z = safety factor against forced rupture; α_k = notch Factor, ν = Poisson's ratio (mostly 0.3); ρ_{mat} = density of impeller material; σ_{bw} = fatigue strength (endurance limit); * = quantity is referred to d_2

Figure 3 [1] - Table 14.1. Impeller or diffuser vane loading

Table 14.2 Impeller shroud loading				Eq.
Alternating pressures acting on impeller shroud				---
Static load on impeller shroud	$\Delta p_{stat} = \frac{\rho}{2} u_2^2 \psi_{Rs}$	$\psi_{Rs} = \frac{2\,(p_{Rs} - p_{La})}{\rho\, u_2^2}$		14.2.1
Dynamic load	$\Delta p_{dyn} = \frac{\rho}{2} u_2^2 \Delta p^* \left(2\frac{x}{t_2} - 1\right)$		$\Delta p^* = \frac{0.008}{(d_3^* - 1)^{0,9}}$	14.2.2
Bending moment M, section modulus W, bending stress σ	$M = \frac{x\,\Delta p\, L_{äq}^2}{60}$	$W = \frac{x}{6} e_{Rs}^2$	$\sigma = \frac{M}{W} = \frac{\Delta p}{10}\left(\frac{L_{äq}}{e_{Rs}}\right)^2$	14.2.3
Impeller shroud thickness e_{Rs}	$e_{Rs} = 1.25\ e$	e = blade thickness (without profile) For e_{Rs} set an average value which is representative for $\Delta r = 5\ e_{Rs}$		14.2.4
Static load on impeller shroud	$\psi_{Rs} = R_G \psi \left\{1 - \frac{r^* - r_1^*}{1 - r_1^*}\right\} - k^2\left(1 - r^{*2}\right) \approx 0.1$			14.2.5
Effective radius (load center)	$r^* = 1 - \frac{2\pi}{3\, z_{La}} \sin\frac{a_2}{t_2} \cos\frac{a_2}{t_2} \approx 0.9$			14.2.6
Equivalent span between two blades	$L_{äq}^* = \frac{L_{äq}}{d_2} = \frac{\pi}{2\, z_{La}}\left(1 + 0.8 \sin\beta_{2B}\right) \approx 0.67\frac{\pi}{z_{La}}$			14.2.7
Mean stress σ_m	$\sigma_m = \frac{\rho}{4} u_2^2 \left(\frac{L_{äq}^*}{e_{Rs}^*}\right)^2 \psi_{Rs}$	$t_2^* = \frac{t_2}{d_2} = \frac{\pi}{z_{La}}$		14.2.8
Alternating bending stress σ_w	$\sigma_w = \alpha_k \frac{\rho}{20} u_2^2 \left(\frac{L_{äq}^*}{e_{Rs}^*}\right)^2 \Delta p^*$			14.2.9
Allowable alternating bending stress $\sigma_{bw,al}$	$\sigma_{bw,al} = \frac{\sigma_{bw}}{S_{bw}}\left(1 - \frac{\sigma_m}{R_m} S_z\right)$			14.2.10
Allowable impeller circumferential speed	$u_{2,al} = 2\,\frac{e_{Rs}^*\, z_{La}}{\pi} \sqrt{\frac{\sigma_{bw}}{\rho\left\{S_{bw}\,\alpha_k \frac{\Delta p^*}{5} + S_z \frac{\sigma_{bw}}{R_m}\psi_{Rs}\right\}}}$			14.2.11

x in Eq (14.2.3) is the width of the "beam" considered; x cancels when calculating the stress.
S_{bw} = safety factor against fatigue fracture; S_Z = safety factor against forced rupture;
α_k = notch factor, ν = Poisson's ratio (mostly 0.3); ρ_{mat} = density of impeller material;
σ_{bw} = fatigue strength (endurance limit); * = quantity is referred to d_2*

Figure 4. [1] – Table 14.2 impeller shroud loading

6. Established Practices

Referring to High Energy Pumps the following limits have been established and they shall be respected regardless of the baseline calculation:

- Head limit at BEP based on maximum continuous speed:
  - Injection Pumps 420 m (1378 ft)
  - Oil Transport Pumps 525 m (1722 ft)
  - Boiler Feed Pumps 700 m (2297 ft)
- Power > 10 MW (13,400 HP)
- Above limits based on increased Gap “B” from 4% to 6%

Vane Thickness	Injection Pumps	Oil Transport Pumps	Boiler Feed Pumps
e/D2= 0.016 (1.6% D2)	> 350 m	> 438 m	> 700 m
e/D2= 0.025 (2.5% D2)	> 420 m	> 525 m	

Table 1. High Energy Pumps established practices.

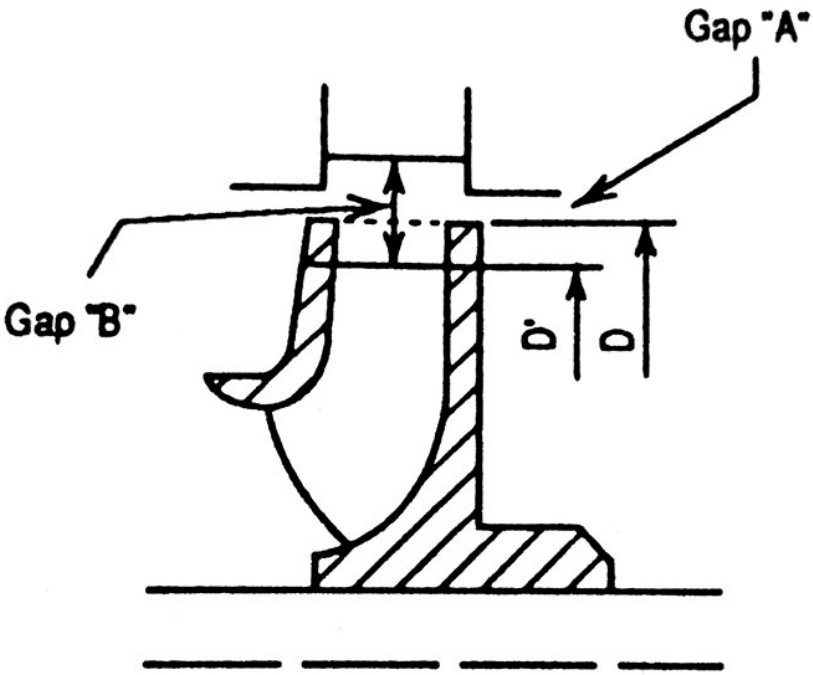


Figure 5. Gap "B" representation.

7. Description of the module

The mask for the input data is shown in Figure 6.

Only data on yellow cells must be compiled.

Data can be filled either on SI or US unit system

PUMP MODEL	
PUMP SIZE	
DRAWING NUMBER	

INPUT DATA at maximum diameter				
unit system	SI		US	
speed	980	RPM	980.0	RPM
Qbep	9800	m3/h	43171.8	GPM
Hstage bep	75	m	22.9	ft
Hstage max	80.0	m	24.4	ft
D2 max	800.0	mm	31.5	in
fluid S.G.	1.0		1.0	
number of vanes Z	7.0		7.0	
channel width between disks (b2 or b3)	87.0	mm	3.4	in
diffuser LE D3 or volute tongue Dcw	850.0	mm	33.5	in
impeller exit blade angle b2b	25.0	°	25.0	°

impeller or diffuser?	impeller
double suction pump?	yes
open or closed?	closed
casting quality	average
sea water fluid?	no

closed impeller		data		
Material		Duplex 3A		
		selected	suggested	
density	ρmat	7700.0	7700	kg/m3
Tensile strength	Ru	650.0	650	MPa
Yeld strength	Rp0.2	480.0	480	MPa
alternate bending strength	σbw	195.0	195	MPa
Safety factor for blade rupture	Sz	2.0	2	
Safety factor for shroud rupture	Szz	5.0	5	

Figure 6: Input data for NPSH Safety margin calculation

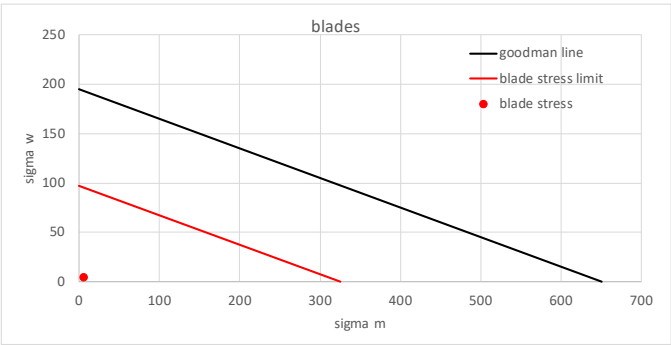
The outcomes of the MODULE are the suggested thickness of impeller or diffuser blade or shroud

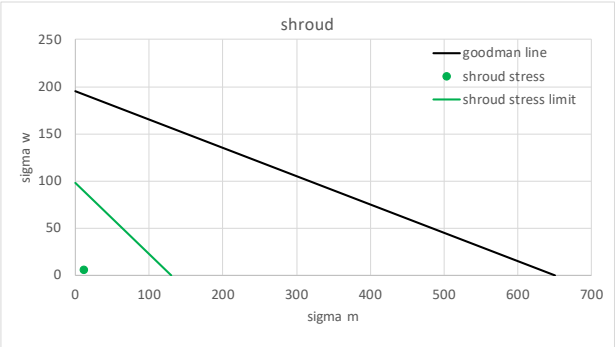
THICKNESS SELECTION AND OUTPUT DATA

		suggested	selected	
nominal thickness	e	13.8	13.50	mm
TE thickness	e2	6.9	10.00	mm
Mean Blade Thickness acc T14.1	em	11.8	12.00	mm
Shroud disk Thickness	eRs	17.2	14.00	mm
fillet radius between blade and disk	rf	4.1	4.00	mm
LE thickness	eLE	8.9		

	u2max	spec speed
SI	41.1	44.9
US	134.7	2316.7

		blades	shroud	
mean stress	σm	6.5	12.4	MPa
alternate bending stress	σw	4.3	4.8	MPa
Safety Factor	SF	15.6	6.9	
Cavitation Erosion Ratio (CA6NM=1, lower is better)			1	





By filling the thickness data on yellow cell allow to determine the stress acting on the blade and shroud component. The safety factor between actual and allowable stress is suggested.

An extra outcome with cavitation erosion ratio of the material (with reference to CA6NM material) is shown. Lower values correspond to better behaviour against cavitation erosion.

8. Limitations

The method is applicable for dimensioning the thickness of impeller and diffuser blades and shrouds considering fatigue stress criteria. No effect due to cavitation erosion have been considered.

In case of special fluid with highly corrosive characteristic special analyses must be carried out by Product Engineering to properly define the idoneus thickness

For minimum thickness calculation the overall dimension of the component, the material characteristics and the castability are primary parameters for the selection of the minimum thickness on blades and shroud disks. Whenever the calculated thickness is below 6 mm, the feasibility of component shall be confirmed by Product Engineering.

9. References

[1] J.F.Gulich “Centrifugal Pumps” Fourth Edition - 2020

10. Attachments

10.3 Check of the blade thickness AXD 800.600 (project 18C039)

As reference, the damage that has occurred during plant operation to the impeller of AXD 800.600 pump is presented here.

As per what discussed in RCA, no single cause of damage has been found, but at the same time several improvements have been identified, to increase the life expectation of the component to acceptable values. The calculation of the blade thickness, for the original design and for the new one and according to this document is described below.

Vaned channels and shroud thickness Calculation

PUMP MODEL	AXD 800.600
PUMP SIZE	
DRAWING NUMBER	

impeller or diffuser?	impeller
double suction pump?	yes
open or closed?	closed
casting quality	average
sea water fluid?	no

INPUT DATA at maximum diameter				
unit system	SI		US	
speed	980	RPM	980.0	RPM
Qbep	9900	m3/h	43612.3	GPM
Hstage bep	75	m	22.7	ft
Hstage max	95.0	m	29.0	ft
D2 max	800.0	mm	31.5	in
fluid S.G.	1.0		1.0	
number of vanes Z	7.0		7.0	
channel width between disks (b2 or b3)	87.0	mm	3.4	in
diffuser LE D3 or volute tongue Dcw	848.0	mm	33.4	in
impeller exit blade angle b2b	25.0	°	25.0	°

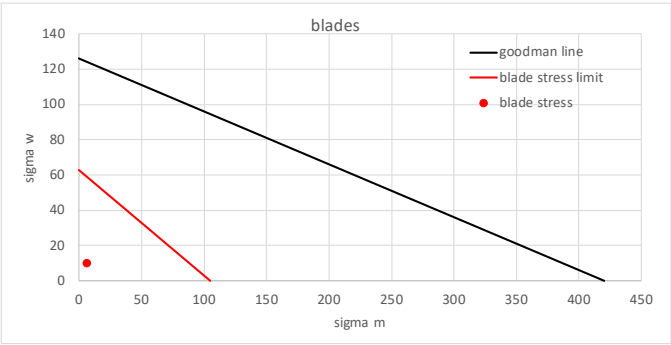
closed impeller		data		
Material		Cast Steel A216 Gr WCB		
		selected	suggested	
density	ρmat	7800.0	7800	kg/m3
Tensile strength	Ru	420.0	420	MPa
Yeld strength	Rp0.2	240.0	240	MPa
alternate bending strength	σbw	126.0	126	MPa
Safety factor for blade rupture	Sz	4.0	4	
Safety factor for shroud rupture	Szz	8.0	8	

THICKNESS SELECTION AND OUTPUT DATA

		suggested	selected	
nominal thickness	e	13.8	9.00	mm
TE thickness	e2	6.9	5.00	mm
Mean Blade Thickness acc T14.1	em	10.3	8.00	mm
Shroud disk Thickness	eRs	17.2	10.00	mm
fillet radius between blade and disk	rf	4.1	5.00	mm
LE thickness	eLE	5.9	5.00	

	u2max	spec speed
SI	41.1	45.3
US	134.7	2340.2

		blades	shroud	
mean stress	σm	6.6	24.4	MPa
alternate bending stress	σw	10.0	9.8	MPa
Safety Factor	SF	4.5	1.6	
Cavitation Erosion Ratio (CA6NM=1, lower is better)			4.7	



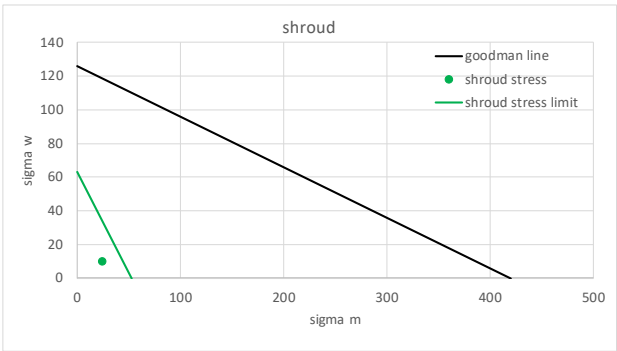


Figure 7.impeller stress safety factor calculation: original configuration

Vaned channels and shroud thickness Calculation

PUMP MODEL	AXD 800.600
PUMP SIZE	
DRAWING NUMBER	

impeller or diffuser?	impeller
double suction pump?	yes
open or closed?	closed
casting quality	average
sea water fluid?	no

INPUT DATA at maximum diameter				
unit system	SI		US	
speed	980	RPM	980.0	RPM
Qbep	9900	m3/h	43612.3	GPM
Hstage bep	75	m	22.7	ft
Hstage max	95.0	m	29.0	ft
D2 max	800.0	mm	31.5	in
fluid S.G.	1.0		1.0	
number of vanes Z	7.0		7.0	
channel width between disks (b2 or b3)	87.0	mm	3.4	in
diffuser LE D3 or volute tongue Dcw	848.0	mm	33.4	in
impeller exit blade angle b2b	25.0	°	25.0	°

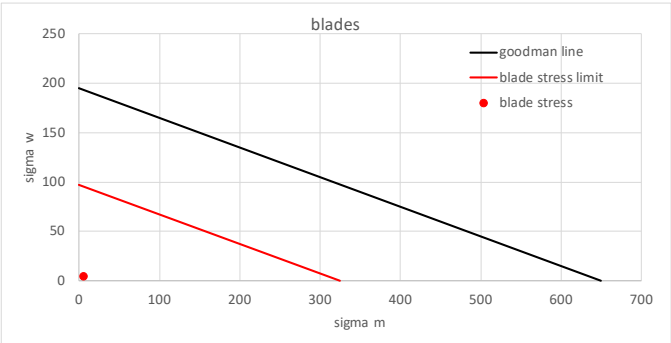
closed impeller		data		
Material		SuperDuplex 6A CD3MWCuN		
		selected	suggested	
density	ρmat	7700.0	7700	kg/m3
Tensile strength	Ru	650.0	650	MPa
Yeld strength	Rp0.2	690.0	690	MPa
alternate bending strength	σbw	195.0	195	MPa
Safety factor for blade rupture	Sz	2.0	2	
Safety factor for shroud rupture	Szz	5.0	5	

THICKNESS SELECTION AND OUTPUT DATA

		suggested	selected	
nominal thickness	e	13.8	13.80	mm
TE thickness	e2	6.9	10.00	mm
Mean Blade Thickness acc T14.1	em	11.9	12.00	mm
Shroud disk Thickness	eRs	17.3	13.00	mm
fillet radius between blade and disk	rf	4.1	5.00	mm
LE thickness	eLE	9.1	9.00	

	u2max	spec speed
SI	41.1	45.3
US	134.7	2340.2

		blades	shroud	
mean stress	σm	5.9	14.4	MPa
alternate bending stress	σw	4.5	5.8	MPa
Safety Factor	SF	15.6	5.9	
Cavitation Erosion Ratio (CA6NM=1, lower is better)			0.9	



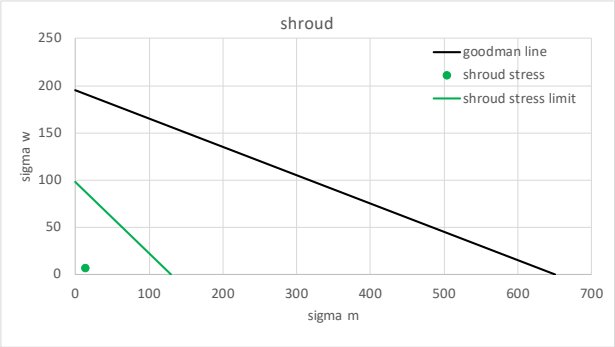


Figure 8 impeller stress safety factor calculation: final redesign

Table 1. comparison of safety factors

	SF-blade	SF-shroud	Cav.Er.Ratio
ORI	4.5	1.6	4.7
NEW	15.6	5.9	0.9

Figure 9 damage of impeller blades at leading edge

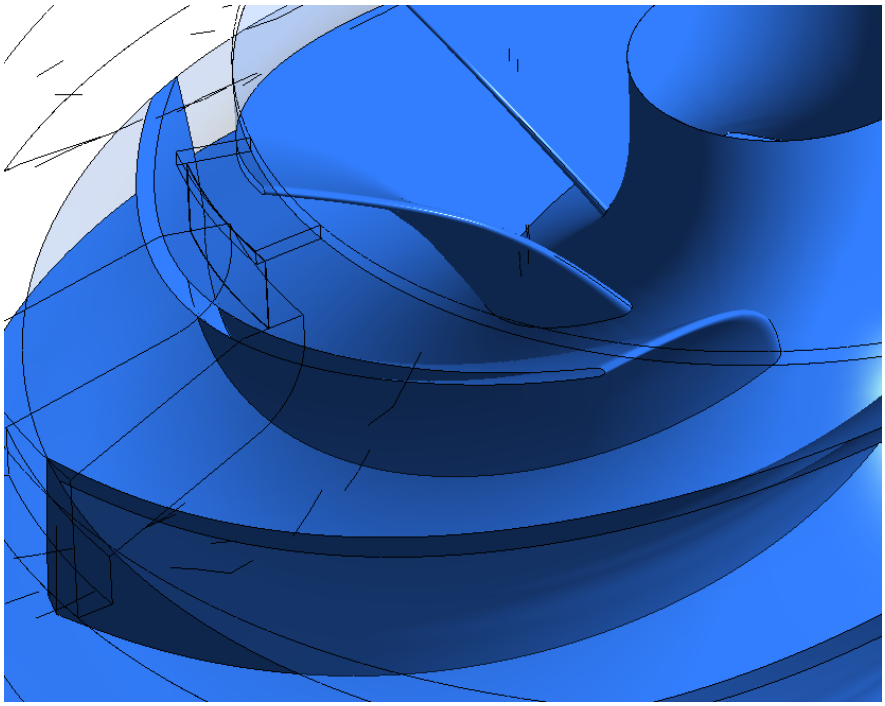


Figure 10 original layout (3D model)

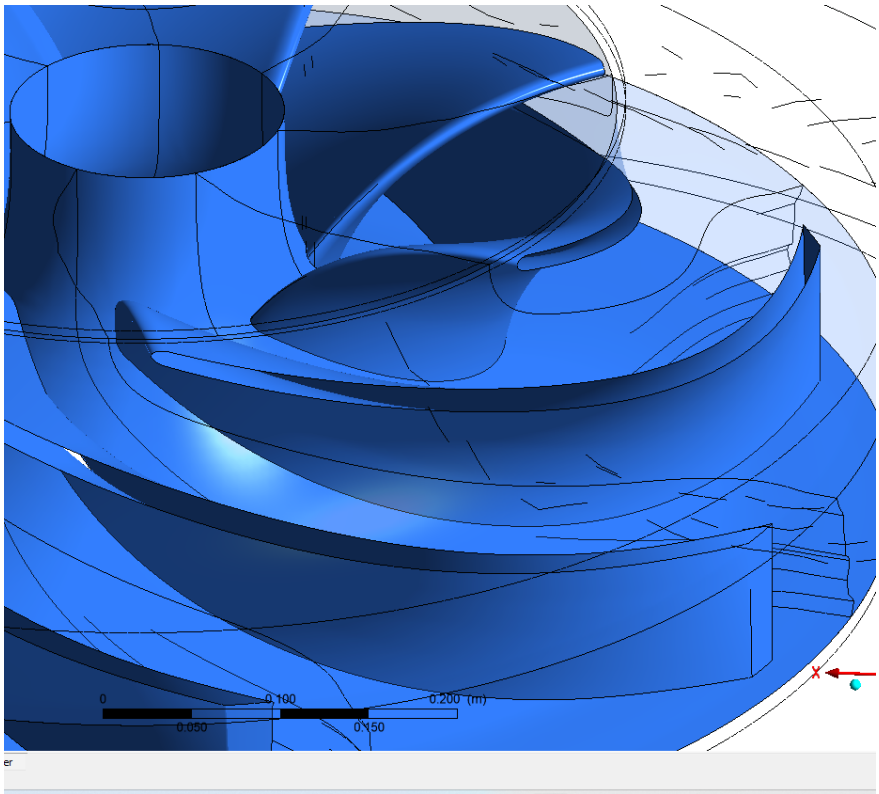


Figure 11 redesign (3D model)